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Course Code: A8707

**VARDHAMAN**

COLLEGE OF ENGINEERING

III B.Tech II Semester Continuous Assessment - I, January - 2026

(Regulations: VCE-R22)

DEEP LEARNING**(Common to CSE , IT)**

Date: 24 January 2026 FN

Time: 2 Hours

Maximum Marks: 30

Answer all Questions in Part-A

Answer any FOUR Questions in Part-B

Course Outcomes with Bloom's Levels:

CO#	CO Statement	Bloom's Level (L#)
CO1	Identify the need of neural networks and deep learning for a given problem.	L02
CO2	Build a CNN model on the real time data.	L04
CO3	Model sequence classification applications using RNN.	L04
CO4	Build a deep learning model using encoders.	L04
CO5	Make use of generative models in various creative tasks	L03

Questions:

PART-A (Multiple Choice / Fill in the Blanks / Match the Following / Short Answer Type Questions)					
			CO#	BL#	Marks
1.	a)	Define Artificial Neural Network and Deep Learning	CO1	BL1	1 M
	b)	What do you understand by a decision surface or decision boundary in a classification problem?	CO1	BL2	1 M
	c)	Discuss the role of hidden units in deep neural networks?	CO1	BL3	1 M
	d)	Enlist the key concepts of Deep Learning	CO1	BL1	1 M
	e)	Define perceptron as single layer feed forward network	CO1	BL1	1 M
	f)	Given an image classification task, why would you choose a CNN over a fully connected neural network?	CO2	BL2	1 M
	g)	What is the effect of increasing the number of filters in a convolutional layer?	CO2	BL1	1 M
	h)	Define Convolutional Neural Network.	CO2	BL1	1 M
	i)	Discuss the role of fully connected layer in CNN	CO2	BL3	1 M
	j)	Differentiate between 1D and 2D with example	CO2	BL2	1 M

PART-B (Descriptive Questions)				
2.	Write short notes on structured outputs and data types in CNNs	CO1	BL2	5 M
3.	Discuss the Gradient Descent learning with example	CO1	BL3	5 M
4.	Construct the architecture of multi-layer feedforward neural network and explain with a neat diagram.	CO1	BL4	5 M
5.	Apply convolution and max pooling operation on the given image represented as a 6 x 6 matrix with a 3 x 3 kernel. $InputImage = \begin{bmatrix} 1 & 2 & 1 & 4 & 3 & 1 \\ 2 & 3 & 4 & 2 & 0 & 9 \\ 1 & 3 & 6 & 7 & 0 & 2 \\ 2 & 3 & 4 & 1 & 7 & 6 \\ 0 & 9 & 0 & 7 & 3 & 7 \\ 7 & 8 & 1 & 3 & 5 & 0 \end{bmatrix}$ $Kernel = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$	CO2	BL4	5 M
6.	Discuss the motivations to use CNN in Deep Learning	CO2	BL3	5 M
7.	Explain various types of pooling and their characteristics with example	CO2	BL2	5 M



VARDHAMAN
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SCHEMA OF VALUATION

Name of the Subject : Deep learning
Programme : B.Tech III-II CSE/IT
Examination : Regular / Supplementary
Max. Marks : 30

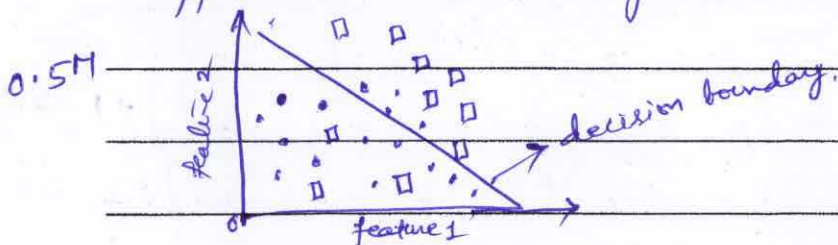
Subject Code : A8707
Year : Jan-2026
Date of Examination : 24-01-2026
Time of Examination : FN

PART - A

Q. 1 a) ANN - Artificial neural network is a computational model inspired by the human brain, consisting of interconnected artificial neurons arranged in layers, which process information by applying weighted sums and activation functions to learn patterns.

DL - Deep learning is a subset of machine learning that uses deep neural networks with multiple hidden layers that automatically learn hierarchical representations of data from large datasets.

b) A decision boundary or decision surface is the boundary in the feature space that separates different classes predicted by a classifier.



c) Hidden neurons or units has two important role:

1) Hidden neurons only receive input from other neurons such as input or other hidden neurons

2) Hidden neurons only output to other neurons, such as output or other hidden neurons.

→ Hidden neurons help the n/w understand the input and form the output. They capture and represent complex ~~part~~ patterns present in the input data.

Q d) Key concepts of Deep learning are:

1. Neural networks.

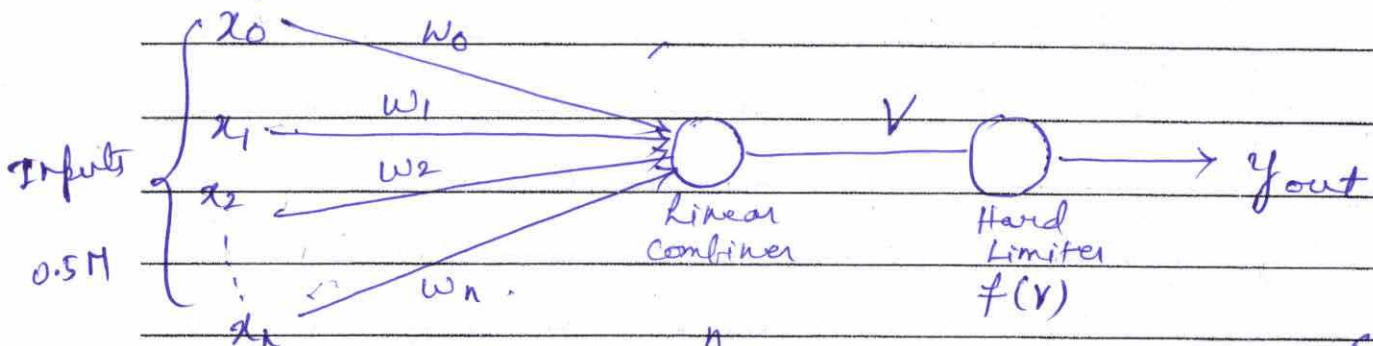
1M 2. Deep ~~Neural~~ neural networks

3. Activation functions

4. Training and Backpropagation.

5. Loss functions.

0.5M © Perceptron is build to overcome the problems of McCulloch-Pitts model. Figure shows the single layer feed forward network.



$$V = \sum_{i=1}^n w_i x_i \quad y_{out} = f(V) = \begin{cases} +1, & V > 0 \\ -1, & V < 0 \end{cases}$$



f) Because CNN's are designed to exploit the spatial structure of images and provide parameter efficiency compare to feedforward fully connected neural network.

→ CNN's provide translation invariance and they are hierarchical feature extraction for better generalization

g) Increasing the number of filters in a convolutional layer effects the network to learn the information during training which improves:

a. Feature learning capacity.

b. Model size.

c. Computational cost

d. Overcome overfitting issue

e. Gradient propagation.

h) A convolutional neural network (CNN) is a specialized type of deep neural network designed to automatically learn and extract hierarchical features from images using convolutional layers applied with filters.

OR

A CNN is one of the main category of deep learning to do image classification and image recognition using neural networks.

l) Role of fully connected layer in CNN

→ The role of fully connected layer acts as a decision-making part of a CNN. The following are the major role of fully connected layer.

1M

1) Feature integration and decision making

2) Flattening of feature maps

3) classification and output mapping

4) Learning non-linear combination

j) Difference between 1D and 2D.

1D (One Dimension)

1) Linear sequence

Ex:- Audio waveforms

2) Single channel

Length x Channel

3) used in speech recognition

2D (Two-Dimension)

1) Grid / matrix (Rows & Columns)

Ex:- Images

2) Multi channel

Height x Width x Channels

3) used for image classification, face recognition.

1M

PART-B

Q.2 → Structured outputs in CNN refer to cases where the NN produces O/P data with a structured or organized format.

→ Structured O/P are common in tasks such as image segmentation, object detection and image captioning.

1M

→ Image Segmentation goal is to partition an input image into multiple segments or regions



→ CNN's can be used to predict a segmentation mask for each pixel in the input image.

→ The o/p of the CNN is a structured grid-like data representing the segmentation mask.

1M 2) Object detection - Object detection involves detecting and localizing multiple objects and classify each bounding box into predefined object classes.

→ The output of the CNN is a structured list of bounding boxes, each associated with a class label and confidence score.

3) Image captioning: - Image captioning tasks involves generating textual descriptions of images.

1M → CNNs can be used to extract features from the input image, which are then fed into a recurrent neural network to generate a descriptive caption.

→ The o/p of CNN-RNN is a structured sequence of words forming a coherent and descriptive caption.

2M → For CNN, the data types used has several channels (single or multi-channels) with different dimensionalities (1-D, 2-D or 3-D).

Q-3) Gradient descent learning is an optimization strategy used in machine learning and deep learning. OR Gradient descent is a optimization algorithm for finding a local minimum of a differentiable function.

3M → The equation below describes the gradient descent algorithm.

$$b = a - \gamma \nabla f(a)$$

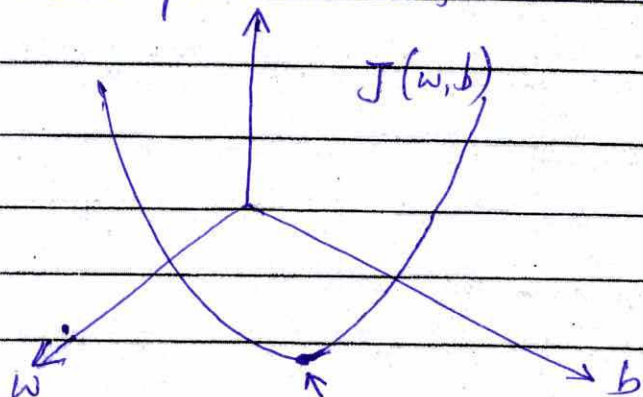
where b is the next position, while ' a ' represents the current position.

→ The minus sign refers to the minimization part of the gradient descent algorithm.

→ The gamma in the middle is a waiting factor and the gradient term $\nabla f(a)$ is simply the direction of the steepest descent.

Example:-

2M → Suppose we have a machine learning problem and want to train the algorithm with gradient descent to minimize the cost function $J(w, b)$ and reach its local minimum by considering the parameters (w and b).



Q.2 continue

(a) 1D Data

* Shape: length $\times$ channels

* Examples: Audio Signals

*

(c) 3D data

* Shape (depth $\times$ height $\times$ width $\times$ channels)

* Examples: Video data & MRI / CT scans

* Model: 3D CNN

(b) 2D Data

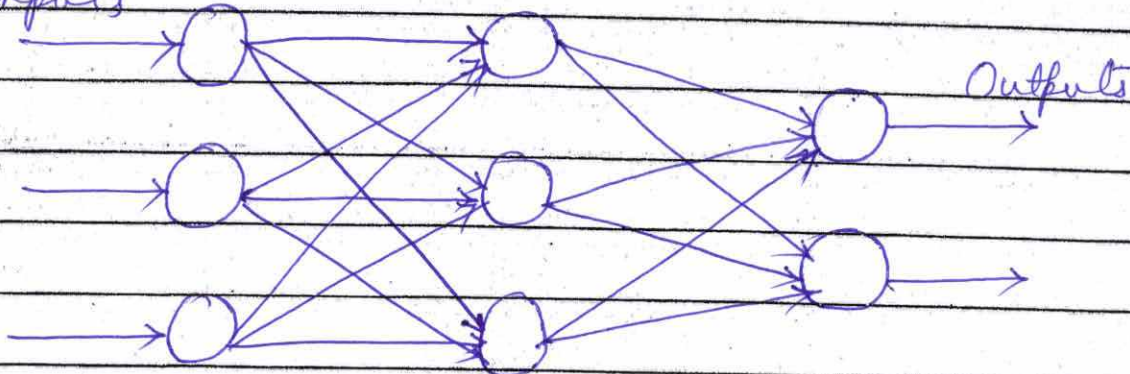
* Shape: (~~depth~~ $\times$ height $\times$ width $\times$ channels)

* Examples: Image, Video data, RGB images, MRI / CT scans

* Model: 2D CNN

Q.4 Architecture of Multi-layer Feed-forward NN.

Inputs



Input layer

Hidden layer

Output layer

→ Input layer - Receives raw input features
- Each neuron represents one feature of the input data

- No computation is performed

→ Hidden layer - one or more layers between input and output.

- Each neuron performs.

$$Z = \sum_{i=1}^n w_i x_i + b$$



$$Y = f(Z)$$

Output layer - produces final prediction

- Number of neurons depends on

* Regression

* Binary classification

* Multi-class classification

Q5)

$$I = \begin{bmatrix} 1 & 2 & 1 & 4 & 3 & 1 \\ 2 & 3 & 4 & 2 & 0 & 9 \\ 1 & 3 & 6 & 7 & 0 & 2 \\ 2 & 3 & 4 & 1 & 7 & 6 \\ 0 & 9 & 0 & 7 & 3 & 7 \\ 7 & 8 & 1 & 3 & 5 & 0 \end{bmatrix}$$

$$K = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

Assume: Stride = 1, No padding (Valid convolution)

Step 1:

$$\text{output size: } (6 - 3 + 1) \times (6 - 3 + 1) \\ = 4 \times 4$$

Step 2:

Convolution Computation

Top left

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 1 & 3 & 6 \end{bmatrix} \odot \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

$$= (1 \times 1) + (2 \times 2) + (1 \times 2) + (2 \times 0) + (3 \times 1) + \\ (4 \times 1) + (1 \times 1) + (3 \times 3) + (6 \times 2)$$

$$= 1 + 4 + 2 + 0 + 3 + 4 + 1 + 9 + 12$$

$$= 36$$

2M

* Repeating the same process for all positions
 Step 3: Final convolution output (feature maps).

3M

$$C' = \begin{bmatrix} 36 & 45 & 37 & 34 \\ 37 & 49 & 46 & 41 \\ 36 & 52 & 41 & 50 \\ 45 & 50 & 37 & 46 \end{bmatrix} //$$

Q. 6) Motivation to CNN leverages three important ideas that help to improve deep learning systems:

2M

1. Sparse interactions.
2. Parameter sharing and
3. Equivariant representations.

1. Sparse interaction:

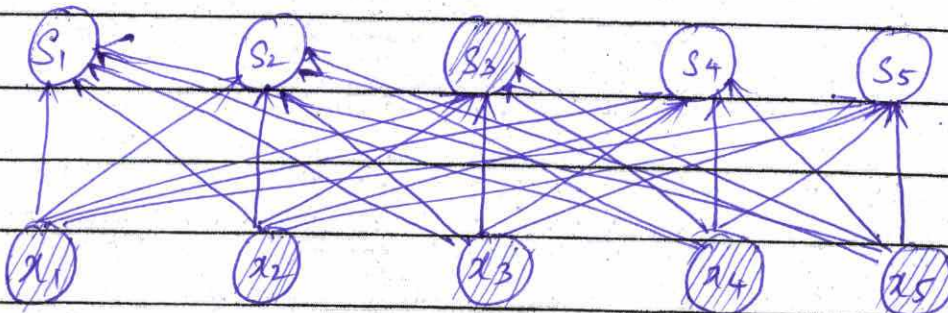
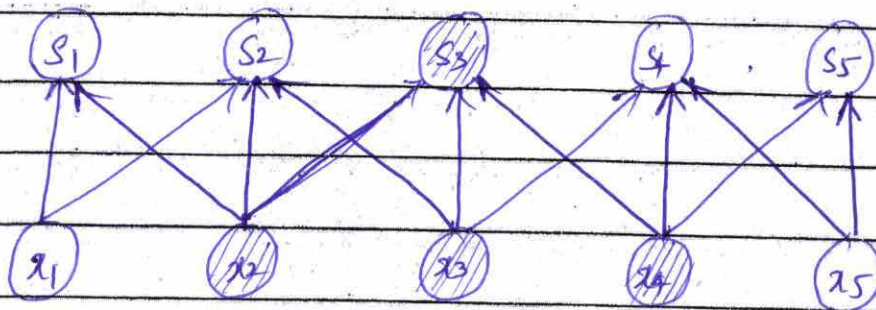
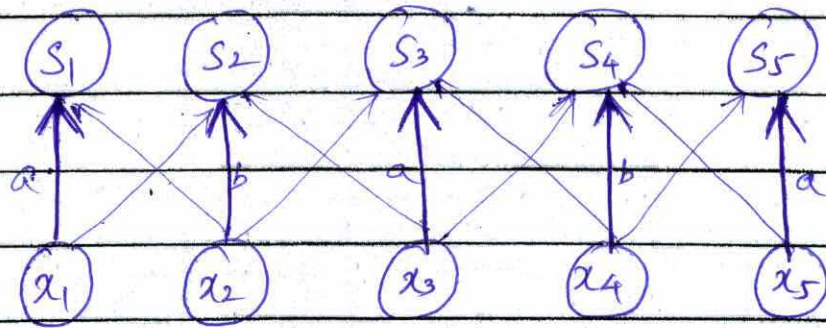


Fig: Sparse connectivity



② Parameter sharing.



Convolution shares the same parameters across all spatial locations.

③ Equivariant representations

- Parameter sharing causes the layer to have a property called equivariance to translation.
- Equivariant means, if the input changes, the output changes in the same way.
- Specifically a function $f(x)$ is equivariant to a function $g(x)$ if

$$f(g(x)) = g(f(x)).$$

Q. 7) Types of pooling in CNNs.

- Pooling pulls most significant features from the convoluted matrix.
- Pooling layer reduces the number of parameters when the image is too large.
- Pooling layer serves two primary purposes.
 - ① Dimensionality reduction.
 - ② Translation invariance.

2M

3M

a) Max-pooling: Computes the maximum value within each local region and helps to capture the most prominent features within each region. By applying a max filter to non-overlapping sub-regions of initial represent.

12	20	30	0
8	12	2	0
34	70	37	4
112	100	25	2

2x2 Max-pool

20	30
112	37

b) Average pooling

4	3	1	5
1	3	4	8
4	5	4	3
6	5	9	4

2x2 Avg. pool

2.8	4.5
5.3	5.6

c) Sum pooling

4	3	1	5
1	3	4	8
4	5	4	3
6	5	9	4

2x2 Sum. pool

11	18
20	20



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